# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Artificial Intelligence and Machine Learning

## ASSESSMENT 3 – Technical Presentation(5 QUESTIONS)

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| **Course Code:** | ITA05 | **Course Title:** | **Computer Vision** |
| **CO Assessed:** | CO1 | **Assessment:** | Assessment 3: Technical Presentation |
| **Weightage:** | 35% of CIA | **Total Marks:** | (5 Questions × 10 Marks) |
| **CO1 Statement:** | ***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.*** | | |
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| **Section / Batch:** | Slot A/ 2024 | **Date:** | 14-10-2026 |

**Code of Conduct**

*I G. Poojitha (192425337) (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: G. Poojitha

**QUESTIONS: 5** **Total Marks: 50**

## Annexure A

## Technical Presentation

**Q1.** A shopping mall uses CCTV cameras that produce clear images during the day but dark and noisy images at night, making it difficult to identify people and objects.
**Prepare a technical presentation demonstrating how filtering, noise removal, contrast enhancement, and sharpening techniques can be applied to improve the captured images. Present the sequence of image-processing operations required before object detection.**

**Q2.** A university is developing an automated document-scanning system using a camera, where captured documents may contain shadows, noise, uneven illumination, tilted pages, and blurred text.
**Prepare a technical presentation explaining how filtering, thresholding, contrast enhancement, edge detection, and segmentation can be applied to improve the quality of the document images. Demonstrate the role of each technique in the processing pipeline.**

**Q3.** A fruit-processing company wants to automatically identify damaged apples from images captured on a conveyor belt, where shadows, reflections, different backgrounds, and brightness variations are present.
**Prepare a technical presentation comparing global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation for this application. Demonstrate how a suitable technique or combination of techniques can improve fruit defect detection.**

**Q4.** An autonomous vehicle uses a front-facing camera to detect lane markings under different weather and lighting conditions, where shadows, damaged markings, vehicles, and rain may affect the captured images.
**Prepare a technical presentation demonstrating how grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and region-of-interest techniques can be applied for lane detection. Explain the challenges associated with each processing stage.**

**Q5.** A municipal corporation wants to classify waste such as plastic, paper, and metal using a camera installed above a conveyor belt. The captured images may contain overlapping objects, shadows, reflections, noise, and different object orientations.
**Prepare a technical presentation demonstrating a complete image-processing pipeline using preprocessing, enhancement, segmentation, feature extraction, and classification. Explain how the proposed pipeline can be applied to handle real-world variations and improve waste classification.**

# Q1. Night CCTV Enhancement

## Problem and Objective

Night-time CCTV footage may be dark, grainy, low in contrast, or blurred. These problems make people and objects difficult to identify and reduce the accuracy of automated object detection. Low illumination means fewer photons reach the sensor; raising camera gain can make sensor noise more visible. The aim is to improve visibility while preserving genuine image details, then provide the processed frame to an object detector.

## Image Formation and Main Defects

A camera records light reflected from objects through a lens onto an image sensor. Recorded pixel intensity depends on illumination, object reflectance, optics, sensor response, and noise. At night, weak illumination lowers pixel values, while moving people may also become blurred during longer exposures. A bright lamp beside a dark corridor creates a large intensity range that one global adjustment may not handle well.

| Defect | Cause | Effect |
|---|---|---|
| Dark image | Insufficient light / exposure | Objects merge with background |
| Grain or speckles | Sensor noise and high gain | False texture and edges |
| Low contrast | Similar object and background intensity | Weak object boundaries |
| Blur | Motion or incorrect focus | Fine details disappear |
| Uneven illumination | Lamps, shadows and reflections | Some areas remain unreadable |

## Engineering Requirements

- Reduce noise without erasing facial, clothing, or object boundaries.
- Improve local contrast without clipping bright lamps or amplifying grain excessively.
- Sharpen only after denoising; strong sharpening creates halos and false edges.
- Retain the original frame and document all processing settings for audit.

A successful solution is not simply the brightest-looking image. It is an image that improves detection reliability across dark corners, bright entrances, moving people, and crowded scenes. Processing should be tested on real footage from each camera because sensor and lighting characteristics differ.

## Filtering and Noise Removal

Filtering reduces unwanted pixel variations before enhancement. A median filter replaces each pixel with the median of neighbouring values and is effective against isolated salt-and-pepper noise. A Gaussian filter computes a weighted average and reduces mild random noise smoothly, but can soften edges. A bilateral filter considers both spatial distance and intensity similarity, smoothing similar regions while preserving many boundaries. Non-local means compares similar patches and can preserve texture, but needs more computation.

| Filter | Strength | Risk / suitable use |
|---|---|---|
| Median | Removes isolated speckles; often preserves edges | Large kernels remove small details |
| Gaussian | Fast and smooth noise reduction | Can blur thin edges |
| Bilateral | Balances smoothing with edge preservation | More computationally expensive |
| Non-local means | Good texture preservation | Higher processing cost |

For real-time CCTV, begin with a small median or Gaussian filter and compare results. Use an edge-preserving method if faces, clothing, or object boundaries become too soft. Avoid strong smoothing: once a small detail is removed, later enhancement cannot reliably recover it.

## Contrast Enhancement

Global histogram equalization redistributes image intensities to improve overall contrast. It is efficient, but may amplify noise and make bright lights harsh. Contrast Limited Adaptive Histogram Equalization (CLAHE) works on local tiles and limits excessive amplification. This can reveal details in shadowed regions without changing the whole frame as aggressively. Apply enhancement to luminance where practical so colour information is not unnecessarily distorted.

## Sharpening

Unsharp masking creates a blurred copy, subtracts it from the original to estimate high-frequency detail, and adds a controlled amount of that detail back. Laplacian-based sharpening also emphasizes rapid intensity changes. Sharpening should follow denoising and contrast enhancement. Too much sharpening makes grain stronger, creates bright/dark halos, and may produce misleading edges. Severe motion blur generally requires improved exposure or specialized restoration, not ordinary sharpening alone.

## Recommended Sequence

**Capture and retain original frame**

↓

**Mild denoising**

↓

**Local contrast enhancement (e.g., CLAHE)**

↓

**Controlled sharpening if needed**

↓

**Normalize image and run object detector**

## Implementation and Tool Selection

A prototype can be built in Python with OpenCV and NumPy. OpenCV supports colour conversion, median/Gaussian/bilateral filtering, CLAHE, and image display. The pipeline should be efficient enough for the required camera frame rate. Store intermediate outputs during testing to determine whether each operation improves the image.

| Stage | Possible tool | Validation check |
|---|---|---|
| Denoising | medianBlur, GaussianBlur, bilateralFilter | Noise falls while boundaries remain clear |
| Contrast | CLAHE; compare histogram equalization | Dark-region detail improves without excessive grain |
| Sharpening | Unsharp mask / Laplacian | Edges improve without halos |
| Detection | Same detector on original and enhanced frames | Recall improves without excessive false positives |

## Evaluation

Use paired original and processed frames containing low-light corridors, bright lamps, moving people, shadows, and crowded areas. Compare both visual quality and model performance. Useful metrics include precision, recall, F1-score, and mean average precision when labelled data are available. Also measure processing time, dropped frames, and stability across cameras. An ablation test—original, denoised, contrast-enhanced, and fully processed—shows which steps contribute measurable benefit.

## Risks and Mitigation

- If brightening reveals too much grain, denoise first and reduce enhancement strength.
- If highlights become clipped, use local contrast adjustment and inspect bright regions.
- If details disappear, reduce filter size and avoid aggressive sharpening.
- If camera conditions vary, tune and validate settings on multiple locations.

## Conclusion

The recommended order is denoising, controlled contrast enhancement, optional sharpening, and object detection. Filtering suppresses distracting noise, CLAHE reveals local detail, and sharpening can improve mild edge softness. Every stage must be tuned using real footage, and originals should be retained for verification. This sequence provides a logical, measurable solution to the stated CCTV problem.

# Q2. Document Image Cleanup

## Problem and Objective

A camera-based document scanner must turn photographs into clean, readable page images for archiving and optical character recognition (OCR). Shadows, noise, uneven illumination, tilted pages, and blur may break letters, merge characters, or make similar symbols difficult to distinguish. The objective is to isolate and straighten the page, create clear text-background separation, and preserve character strokes for OCR.

## Sources of Degradation

| Issue | Effect on image | Possible OCR impact |
|---|---|---|
| Shadows / uneven light | One page region is darker | Faint text disappears after thresholding |
| Noise | Dots and random intensity changes | False punctuation or character fragments |
| Tilt / perspective | Slanted lines or trapezoid page | Incorrect line order and segmentation |
| Blur | Strokes merge or fade | Similar letters are confused |
| Busy background | Desk patterns remain visible | Page boundary and text are misidentified |

## Processing Pipeline

**Capture document image**

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**Detect page and correct tilt / perspective**

↓

**Convert to grayscale or luminance**

↓

**Denoise and correct illumination**

↓

**Enhance contrast**

↓

**Detect page / structural edges**

↓

**Threshold foreground text**

↓

**Segment layout and text regions**

↓

**Run OCR and inspect output**

## Filtering

A median filter can remove isolated speckles; a small Gaussian filter can reduce mild random noise. Filtering should be conservative because thin character strokes, punctuation, and dots are meaningful. Denoising should occur before thresholding: thresholding converts grey values into binary decisions, and a faint stroke removed at this stage is difficult to reconstruct later.

## Geometry Correction

The page boundary can be estimated from contours or edges. If the page is tilted, rotate it to align text lines. If perspective makes the page trapezoidal, estimate its four corners and apply a perspective transform. Crop only after checking that all page content remains inside the corrected boundary. Severe blur is best addressed by recapturing with correct focus rather than relying on enhancement alone.

## Illumination and Contrast Enhancement

Contrast enhancement makes printed strokes more distinct from the page background. Global histogram equalization can help when the whole image has low contrast. For shadows and local brightness changes, CLAHE or background-illumination normalization is often more appropriate. Enhancement must not turn paper texture into noise or clip faint grey text into white.

## Thresholding

Thresholding converts grayscale pixels into foreground text and background. Global thresholding applies one value to the whole image and works best under uniform lighting. Otsu's method estimates a threshold from the intensity histogram. Adaptive thresholding calculates a local threshold and is usually more robust when shadows or illumination vary across the page.

| Method | Best condition | Limitation |
|---|---|---|
| Global | Uniform lighting and clear separation | Fails when one value cannot separate all text |
| Otsu | Histogram separates foreground and background | Struggles with strong illumination gradients |
| Adaptive | Shadows and local brightness changes | May amplify texture; neighbourhood size matters |

After thresholding, inspect whether thin strokes have broken, letter holes have filled, or speckles remain. Morphological opening can remove tiny foreground noise; closing can connect small gaps. These operations must be modest because they can alter punctuation and glyph shape.

## Edge Detection

Sobel filters estimate intensity gradients; Canny detection smooths the image, finds gradients, thins edges, and links edge segments using two thresholds. Edges can locate page boundaries, table lines, text blocks, and other structures. However, every printed character contains edges, and background texture can create false ones. Edge maps guide geometry and segmentation but do not replace a clean text image.

## Segmentation

Segmentation divides the page into meaningful regions. Page segmentation removes the desk background; layout segmentation identifies columns, tables, images, and paragraphs; line or word segmentation prepares regions for OCR. Connected-component analysis groups neighbouring foreground pixels, but touching characters and punctuation require careful handling.

## Tools and Practical Workflow

| Task | Possible method | Purpose |
|---|---|---|
| Grayscale | OpenCV colour conversion | Simplify intensity processing |
| Denoise | Median / Gaussian filter | Reduce speckles and random variation |
| Geometry | Contours, line detection, perspective transform | Straighten and crop the page |
| Enhancement | CLAHE / background normalization | Improve text visibility |
| Threshold | Global, Otsu, adaptive | Separate text from paper |
| Edges / segmentation | Sobel, Canny, contours, components | Locate boundaries and text regions |
| Recognition | OCR engine | Convert page image to editable text |

## Validation

Test bright and dim rooms, side shadows, tilted pages, coloured paper, small fonts, tables, and mildly blurred images. Compare recognized text with a manually checked reference. Character error rate and word error rate measure OCR mistakes. Also check page-crop accuracy, skew correction, processing time, and whether the binary image retains punctuation and thin strokes.

## Common Failures

- Residual shadows: normalize background illumination or use adaptive thresholding.
- Missing strokes: reduce smoothing and tune the threshold.
- Desk texture: improve page detection and exclude the background.
- Cropped content: verify corner estimates and retain a small safety margin.
- Unreadable blur: recapture with better focus where possible.

## Conclusion

A reliable scanner combines page detection and geometry correction, filtering, illumination correction, contrast enhancement, edge detection, thresholding, and hierarchical segmentation. Each stage has a separate purpose: filtering removes noise, enhancement improves visibility, edges locate structure, thresholding separates text, and segmentation organizes content for OCR. Test the complete pipeline using image quality and recognition accuracy rather than appearance alone.

# Q3. Apple Defect Segmentation

## Problem and Objective

A fruit-processing company needs to identify damaged apples on a conveyor belt. Shadows, reflections, varying backgrounds, and brightness changes can make healthy skin resemble bruises or rot. The system must first isolate each apple from the belt and then identify suspicious surface regions. A dark patch alone is not sufficient evidence of damage because natural skin colour, stem marks, and lighting can produce similar appearances.

## Global Thresholding

Global thresholding classifies pixels using one threshold for the whole image. It is fast and simple, making it useful when lighting, background, and apple appearance are controlled. It can create a candidate mask for a dark defect against lighter skin. Its weakness is that shadows and bright reflections shift pixel intensity: one threshold may label a healthy shadow as a defect while missing a defect hidden by glare.

## Adaptive Thresholding

Adaptive thresholding calculates a local threshold from a neighbourhood, such as a local mean or weighted mean. It can handle uneven illumination better than a single global value. However, a small neighbourhood may react to pores and skin texture; a large neighbourhood may miss small local changes. Reflections and naturally dark patches can still produce false candidates.

| Method | Advantage | Limitation |
|---|---|---|
| Global threshold | Fast and simple in stable conditions | Sensitive to global brightness changes |
| Adaptive threshold | Handles local illumination variation | Can misclassify pores, texture, and reflections |

## Why Thresholding Alone Is Not Enough

Defect appearance depends on apple variety, ripeness, defect age, camera exposure, and viewing angle. Thresholding should therefore generate candidate regions, not make the final decision. A reliable solution combines segmentation with colour, texture, shape, and contextual features. Controlled diffuse lighting and a consistent camera position reduce the variation that software must handle.

## Edge-Based Segmentation

Edge-based segmentation locates boundaries where intensity or colour changes rapidly. Sobel estimates gradients, while Canny produces thin contours after smoothing, non-maximum suppression, and hysteresis thresholding. The outer apple contour can separate fruit from the belt; strong internal edges may outline cuts or sharply bounded bruises.

Healthy skin can also contain strong edges from stems, reflections, and natural markings. A soft bruise may have no clear boundary, and noise may break contours. Edge-based segmentation is best used with contour validation, local colour evidence, and region refinement rather than as the only method.

## Region-Based Segmentation

Region-based methods group pixels with similar colour, intensity, or texture. Region growing expands from seed pixels; clustering methods such as k-means group pixels by feature similarity; marker-based watershed can separate touching objects when markers are reliable. These methods can form coherent defect patches instead of isolated edge fragments.

Their limitations include splitting healthy skin into several regions because of natural colour variation, or treating a highlight as a separate region. Results depend on colour space, seed selection, similarity thresholds, and post-processing.

| Technique | Role | Risk and response |
|---|---|---|
| Canny / gradients | Locate outer contour and sharp defect edges | Broken or irrelevant edges; validate shape |
| Region growing | Expand candidate defect from seed | Seed-sensitive; use size limits |
| Colour clustering | Group fruit, belt, and unusual patches | Colour overlap; combine with texture |
| Watershed | Separate touching apples / adjacent regions | Over-segmentation; use markers |

## Combined Pipeline

**Capture under diffuse, stable lighting**

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**Denoise and normalize illumination**

↓

**Segment apple from conveyor**

↓

**Separate touching fruit if needed**

↓

**Create defect candidates with local thresholding**

↓

**Refine using edges and region similarity**

↓

**Extract features and classify**

## Feature Extraction and Classification

| Feature | Examples | Purpose |
|---|---|---|
| Colour | Mean colour; difference from nearby healthy skin | Detect unusual dark or brown patches |
| Texture | Local variance, gradient statistics, local binary patterns | Distinguish damaged tissue from smooth reflections |
| Geometry | Area, perimeter, circularity, aspect ratio | Reject tiny speckles and implausible shapes |
| Context | Position near stem; surrounding region | Avoid confusing expected structures with defects |
| Learned features | CNN features from labelled patches | Learn complex combinations when data are sufficient |

A rule-based classifier can be a baseline in a tightly controlled environment. A supervised classifier or convolutional neural network can generalize better when trained on diverse labelled examples. Include healthy apples, bruises, cuts, rot, shadows, reflections, varieties, and defect severities. Low-confidence cases should be sent for manual review rather than forced into a confident label.

## Evaluation Plan

- Measure precision, recall, F1-score, and a confusion matrix for defect classification.
- Measure segmentation quality with intersection over union (IoU) or Dice score against labelled masks.
- Track false rejection of healthy apples and false acceptance of damaged apples.
- Split data by capture session or batch to reduce leakage between similar images.

## Real-World Mitigation

Use diffuse lighting to reduce specular reflection, local illumination correction for shadows, and a controlled conveyor background. Space apples apart where possible. For overlapping fruit, consider instance segmentation or marker-based watershed. Use adequate image resolution for small defects and update the dataset when new varieties or packaging conditions appear.

## Conclusion

Global thresholding is efficient under stable lighting; adaptive thresholding handles local brightness changes; edge methods locate boundaries; and region methods form coherent patches. Combining them with colour, texture, geometry, and classification is more reliable than using any one technique. Validate segmentation and defect decisions separately to reveal where errors occur.

# Q4. Autonomous Lane Detection

## Problem and Objective

An autonomous vehicle uses a front-facing camera to locate lane markings under changing weather and lighting. Shadows, rain, wet-road reflections, faded paint, damaged markings, vehicles, and road repairs may create false or missing visual cues. The objective is to estimate plausible lane boundaries and provide a confidence value; the system should acknowledge uncertainty when evidence is weak.

## Grayscale Conversion

Grayscale conversion reduces colour pixels to intensity values, simplifying gradient and threshold operations. It works when lane paint is brighter than the road. However, colour can distinguish yellow and white markings and may help reject certain reflections. A practical system retains the original colour image while deriving grayscale for edge processing.

## Filtering

A small Gaussian filter reduces sensor noise and road texture before edge detection; a median filter can remove isolated speckles. Excessive smoothing can erase thin or worn lane markings. Filter size must be tuned for the camera resolution and checked on both clear and faded paint.

## Contrast Enhancement

CLAHE can improve local contrast in shadowed road regions, while gamma adjustment changes overall brightness in a controlled way. Enhancement may also make road cracks, water reflections, and vehicle edges stronger. The aim is to improve lane evidence without increasing false candidates.

| Stage | Purpose | Challenge |
|---|---|---|
| Grayscale / colour | Simplify intensity processing while preserving colour cues | Brightness alone may not distinguish paint from glare |
| Filtering | Reduce noise before gradients | Over-smoothing erases faded markings |
| Contrast | Reveal low-contrast lane paint | May amplify road texture and reflections |

## Edge Detection

Canny edge detection smooths the image, estimates gradients, thins edges through non-maximum suppression, and links edges with two thresholds. Lane markings often form elongated boundaries converging toward the horizon. Vehicles, curbs, road seams, puddles, and shadows also create edges, so the edge map must be filtered using direction, continuity, and plausible lane geometry.

## Thresholding

Thresholding selects pixels likely to be lane paint. A brightness threshold may isolate white markings; colour thresholds in a suitable colour space may identify yellow paint. Adaptive thresholding can handle local

illumination variation, but it may also select bright texture and reflections. Combine thresholds with shape and position constraints instead of treating every bright pixel as a lane.

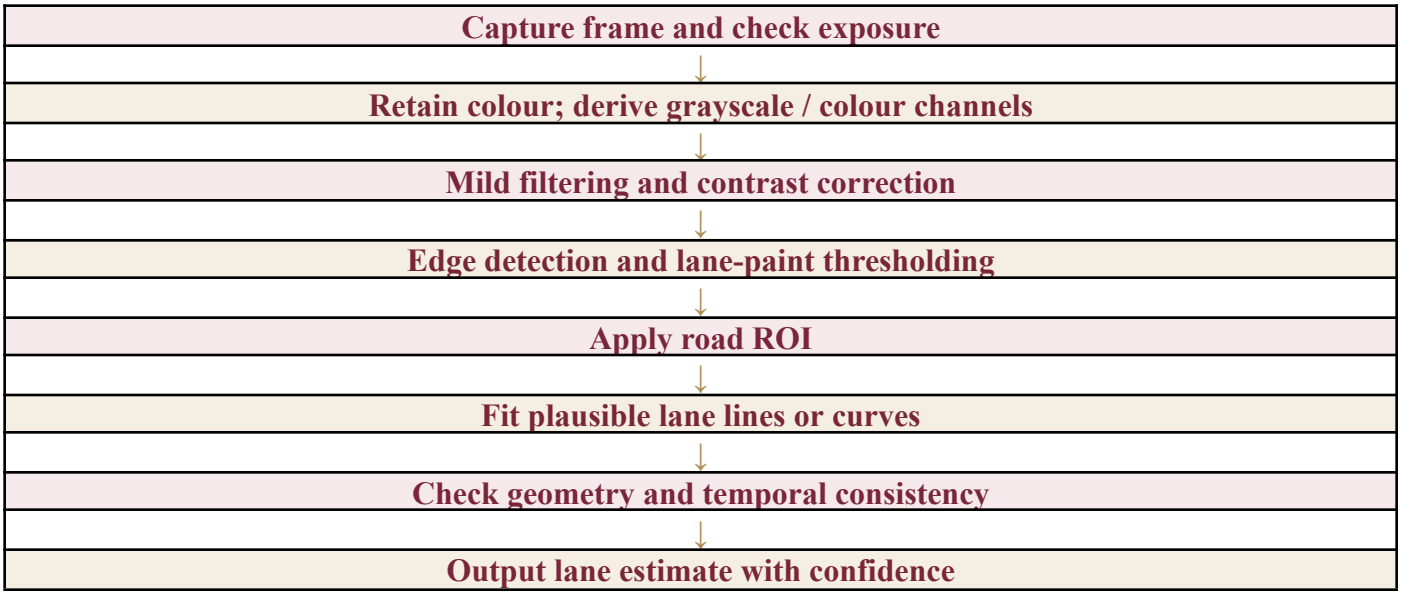
## Region of Interest (ROI)

An ROI restricts processing to the road area where lanes are expected, often a polygon that excludes sky and roadside buildings. This reduces irrelevant edges and computation. A fixed ROI can fail on sharp curves, hills, intersections, or lane changes, so it should be calibrated and, where necessary, adjusted dynamically.

## Lane Fitting

Candidate pixels can be fitted with straight lines on simple roads or curves/polynomials when the road bends. Robust fitting reduces the influence of outliers. Estimated lane width, curvature, and changes between frames should remain plausible. If one lane marking is hidden, the system may use prior frames cautiously, with reduced confidence.

## Processing Pipeline




## Stage-Specific Challenges

| Condition | Failure mode | Mitigation |
|---|---|---|
| Shadows | Shadow boundary looks like a lane | Use colour, local normalization, temporal checks |
| Rain / wet road | Reflections create bright candidates | Use geometry and confidence checks |
| Faded paint | Edges and masks are fragmented | Allow limited gaps and track across frames |
| Vehicles | Lane pixels are occluded | Use visible segments and temporal tracking |
| Curves / hills | Fixed ROI or straight-line model fails | Dynamic ROI and curve fitting |
| Night / glare | Low signal and large brightness range | Exposure control and conservative enhancement |

## Evaluation and Safety

Test on day and night roads, wet and dry surfaces, shadows, worn markings, construction zones, curves, and partial occlusions. Use annotated lane boundaries and held-out routes. Measure boundary error, false positives, missed lanes, temporal jitter, availability, and performance by weather condition. An overall average may hide dangerous failures in a specific scenario.

- Monitor confidence and define a safe fallback when image evidence is weak.
- Do not treat a camera-only lane detector as sufficient for every driving decision.
- Check performance at realistic frame rates and processing latency.

## Conclusion

A practical lane pipeline retains colour cues, derives grayscale, filters lightly, enhances contrast, detects edges, thresholds likely paint, masks an ROI, and fits lane geometry. Each stage has limitations: filtering may erase paint, enhancement may amplify glare, edges may select shadows, thresholds may fail in rain, and fixed ROIs may miss curves. Combining cues, tracking over time, and validating under varied conditions improves robustness.

# Q5. Waste Image Classification

## Problem and Objective

A municipal corporation wants to classify plastic, paper, and metal using a camera above a conveyor belt. Overlapping objects, shadows, reflections, noise, and different orientations complicate recognition. The system must isolate each item, identify useful visual features, assign a category, and send a reliable result to the sorting stage. Detection/segmentation and classification are separate tasks: a classifier cannot reliably label two items that have merged into one crop.

## Sources of Variation

| Variation | Effect | Response |
|---|---|---|
| Shadow / uneven light | Same material has different intensity | Normalize illumination and use multiple features |
| Reflection | Surface detail is hidden | Diffuse lighting and reflection examples in training |
| Noise | False small regions | Mild denoising |
| Orientation | Shape changes in the image | Rotation-tolerant features / varied training data |
| Overlap | Objects merge into one region | Instance segmentation or spread items apart |
| Dirt / crushing | Appearance differs from clean examples | Include realistic training samples |

## Preprocessing

Stable camera position, correct focus, suitable shutter speed, and diffuse lighting improve the input before software begins. A conveyor surface that contrasts with common waste makes foreground segmentation easier. Preprocessing can include resizing, colour-space conversion, mild denoising, and illumination correction. Gaussian filtering reduces random noise; median filtering removes isolated speckles. Strong smoothing must be avoided because thin paper edges and small metal pieces may disappear.

## Enhancement

CLAHE can improve local contrast, while brightness or colour normalization reduces lighting variation. However, material colour is often useful, so enhancement must not distort it excessively. Keep the original image and compare it with the enhanced branch to ensure that useful material cues remain visible.

## Segmentation

Segmentation separates waste from the conveyor and ideally isolates each item. Background subtraction works when the belt is stable. Thresholding can isolate objects that contrast with the belt; edge methods find boundaries; region methods group similar pixels. Transparent plastic may have weak boundaries, shiny metal can blend with highlights, white paper may merge with a pale belt, and overlapping objects may form one connected region.

| Method | Useful condition | Limitation |
|---|---|---|
| Background subtraction | Stable, known belt | Sensitive to belt and shadow changes |
| Threshold / colour mask | Clear object-background difference | Fails when colours overlap |
| Edges / contours | Visible boundaries | Broken edges and overlap cause ambiguity |
| Region / clustering | Coherent colour or texture groups | One item may split or items may merge |
| Instance segmentation | Touching or overlapping objects | Needs labelled data and more computation |

## Handling Overlap

Connected-component analysis may treat touching items as one object. A distance transform and marker-based watershed can sometimes split joined shapes when centres and boundaries are distinguishable. For irregular or heavily overlapping waste, instance-segmentation models are more flexible. Physical spacing of items on the conveyor can reduce the difficulty before image processing.

## Feature Extraction

| Feature | Examples | Value |
|---|---|---|
| Colour | Mean and distribution of channels | Useful for coloured plastics and paper |
| Texture | Local binary patterns, gradient histograms | Describes printed, rough, or smooth surfaces |
| Shape | Area, perimeter, aspect ratio, contour | Helps distinguish flat paper, bottles, cans |
| Edges | Edge density and contour structure | Captures folds and rigid boundaries |
| Learned features | CNN image embeddings | Learns complex patterns from examples |

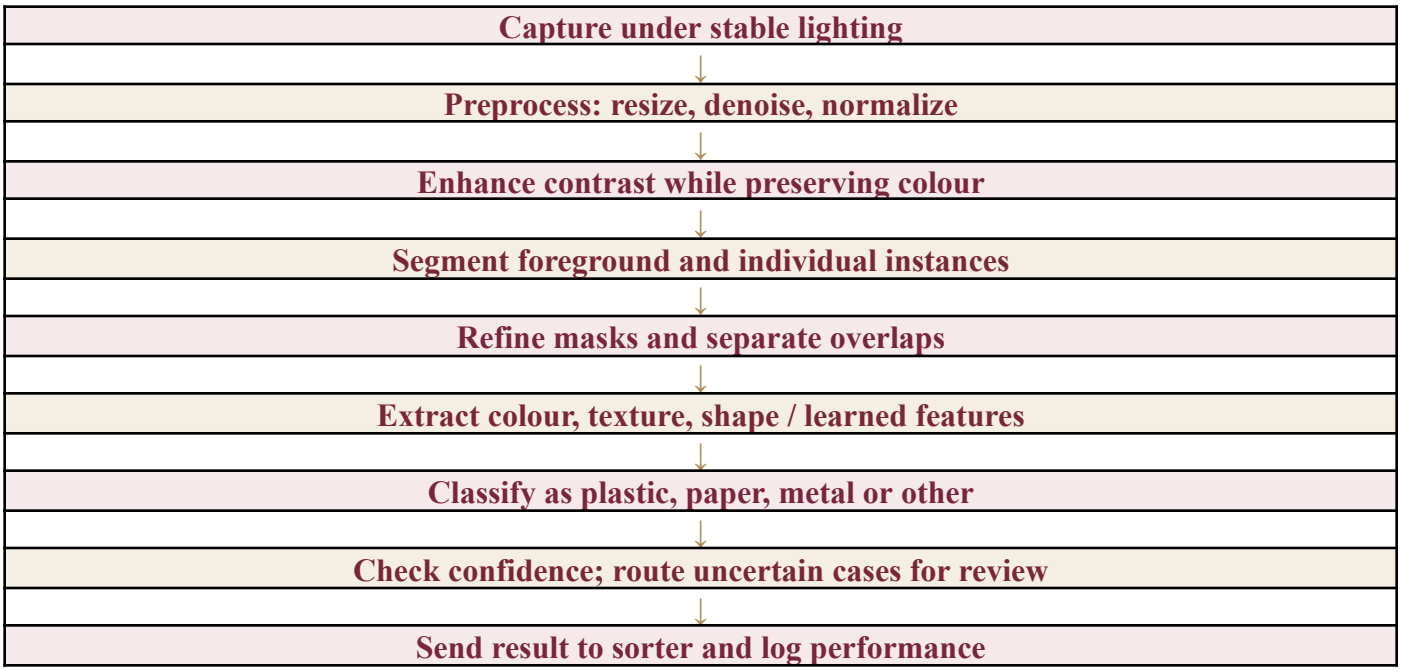
Traditional machine-learning systems use engineered features; deep-learning models learn features from labelled images. Both require representative training examples and separate evaluation.

## Classification

For a small controlled dataset, support vector machines, random forests, or k-nearest neighbours can form useful baselines when paired with selected features. A convolutional neural network may perform better with a larger, diverse dataset. Include varied orientations, sizes, lighting, dirt, crushing, labels, and partial occlusions. Split data by collection session or site to test generalization rather than relying on near-duplicate frames.

The system should provide a confidence score and allow an ‘other/unknown’ or manual-review outcome. Forcing a label on an uncertain object can send material to the wrong bin. The category result should remain linked to the item’s conveyor location so the sorting mechanism acts at the correct time.

## Complete Pipeline




## Evaluation

| Measure | Meaning |
|---|---|
| Precision / recall per class | Correctness and missed items by category |
| F1-score / confusion matrix | Balance and category confusions |
| IoU / Dice | Quality of segmented object masks |
| Latency / throughput | Whether processing matches belt speed |
| Sorting error rate | Practical effect of predictions |
| Manual-review rate | Frequency of uncertain decisions |

## Deployment and Conclusion

Calibrate the camera, lighting, belt, and sorting mechanism together. Account for conveyor speed and processing delay when tracking an item to the sorter. Monitor performance because waste streams change over time. Log model version, confidence, and failures; periodically review new examples and retrain in a controlled manner.

The complete solution uses preprocessing, enhancement, segmentation, feature extraction, and classification. Real-world robustness comes from handling overlap and reflection through both engineering controls and diverse data. Evaluate segmentation and classification separately and together, using accuracy, latency, and actual sorting errors.

## Rubrics for Calculation

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Technical Content Accuracy** | 30 | Highly accurate, indepth, and insightful technical explanation | Technically correct content with adequate depth and clarity | Basic concepts presented with limited accuracy and depth | Content contains major technical errors; concepts poorly understood |
| **Problem Identification** | 25 | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach | Problem identified but analysis or methodology is weak | Problem statement unclear or incorrect; no logical approach |
| **Use of Tools** | 20 | Advanced tools, well-analyzed data, and highly effective visuals | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation | Minimal or incorrect use of tools and data; poor visuals |
| **Communication** | 15 | Highly engaging, confident, and audience-focused delivery | Clear, organized, and professional communication | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides |
| **Professionalism** | 10 | Exemplary professionalism, leadership, and ethical awareness | Professional conduct with effective team collaboration | Basic professionalism; uneven team participation | Lacks professionalism; minimal contribution or ethical awareness |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| Signature: _____<br>Date: _____ | Signature: _____<br>Date: _____ | Signature: _____<br>Date: _____ |